Numerical Methods for the Jin-Xin Model

IMEX Schemes and RelaxNN Frameworks

Stefano Magrini Alunno (*candidate*)

Kinetic Equations: Classical Foundations and Modern Computational Perspectives Examination (July 23, 2026)

Department of Mathematics and Computer Science - University of Ferrara

Outline

Introduction & Mathematical Background

Implicit-Explicit Methods & Relax Neural Networks

Implicit Explicit Neural Network (Proposal)

References

[1] S. Jin and Z. Xin, **"The relaxation schemes for systems of conservation laws in arbitrary space dimensions,"** *Communications on Pure and Applied Mathematics*, 1995.

[2] U. M. Ascher, S. J. Ruuth, and R. J. Spiteri, **"Implicit-explicit runge-kutta methods for time-dependent partial differential equations,"** *Applied Numerical Mathematics*, 1997.

[3] S. Boscarino, L. Pareschi, and G. Russo, *Implicit-Explicit Methods for Evolutionary Partial Differential Equations*. Society for Industrial and Applied Mathematics, 2024.

[4] N. Zhou and Z. Ma, ***Capturing shock waves by relaxation neural networks*****,** 2024.

Introduction & Mathematical Background

The Target & Jin-Xin Relaxation Approximation

Target: Viscous/Inviscid Burgers' Equation

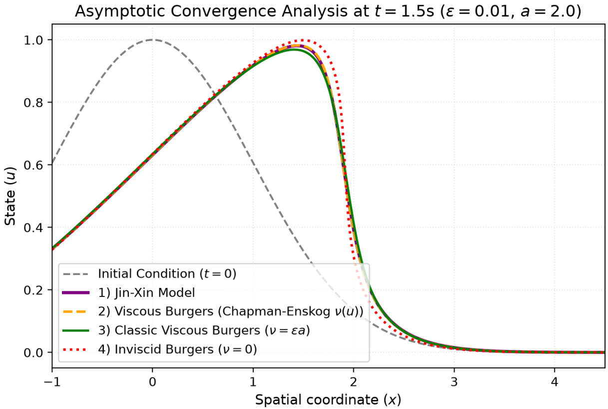
$$\partial_t u + \partial_x \left(f(u)\right) = 0 \quad \xrightarrow{\text{viscous reg.}} \quad \partial_t u + u\partial_x u = \nu \partial_{xx} u$$

Linear Hyperbolic System with Stiff Relaxation (Jin-Xin)

$$\begin{cases} \partial_t u + \partial_x v = 0 \\ \partial_t v + a\partial_x u = -\frac{1}{\varepsilon}(v - f(u)) \end{cases}$$

Key Properties

- **Characteristic Speeds**: Fixed linear velocities $\pm\sqrt{a}$ (Riemann Invariants).
- **Sub-characteristic Condition**: $a > (f'(u))^2$ guarantees linear stability.



Chapman-Enskog ($\varepsilon \to 0$) : $\quad \partial_t u + \partial_x f(u) = \partial_x \left[\varepsilon \left(a - \left(f'(u)\right)^2\right) \partial_x u\right] + \mathcal{O}(\varepsilon^2)$

Implicit-Explicit Methods & Relax Neural Networks

Numerical Solvers vs. SciML: The High-Dimensional Challenge

$$\begin{cases} \partial_t u + \nabla_x \cdot \mathbf{v} = 0 \\ \partial_t \mathbf{v} + a \nabla_x u = -\frac{1}{\varepsilon} \left(\mathbf{v} - \mathbf{f}(u) \right) \end{cases}$$

- **Classical Bottleneck**: Finite Volume & IMEX scale poorly in high-dimensional settings (*Curse of Dimensionality*) and are bounded by mesh CFL constraints.
- **SciML Paradigm Shift**: Neural Operators act as **mesh-free, continuous maps**. Inference is $\mathcal{O}(1)$, invariant to spatial resolution or parameters (ε, a).

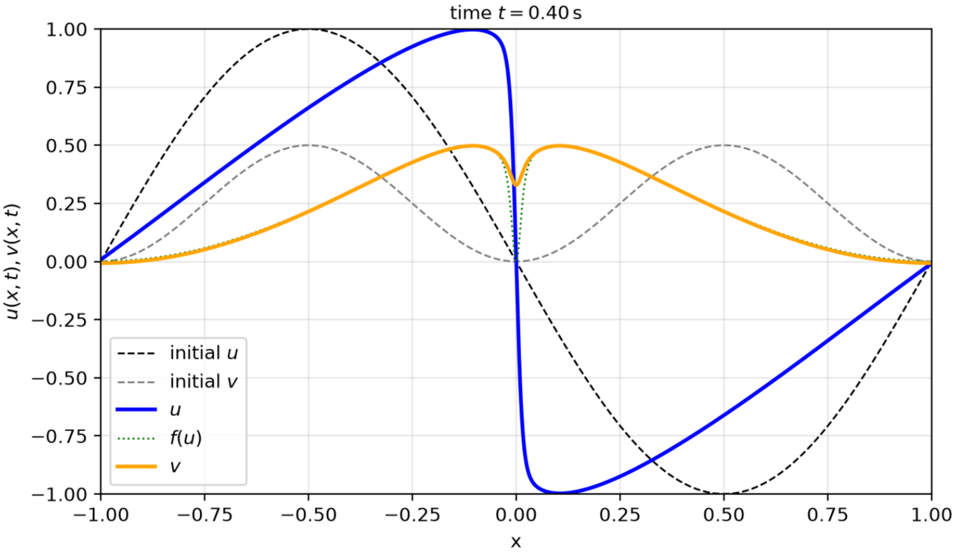
Operator Splitting for Multi-Scale Systems ($\varepsilon \ll 1$)

- **Explicit Step (Convection)**: Resolve linear advection via Rusanov flux:

$$u^{n+1} = u^n - \Delta t \, \nabla_x \cdot \mathbf{v}^n$$

- **Implicit Step (Relaxation)**: Treat stiff algebraic source implicitly:

$$\mathbf{v}^{n+1} = \frac{\varepsilon \mathbf{v}^n - \varepsilon \Delta t a \nabla_x u^n + \Delta t \mathbf{f}\left(u^{n+1}\right)}{\varepsilon + \Delta t}$$



time $t = 0.40\,\mathrm{s}$
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
$u(x,t), v(x,t)$
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
x
initial u
initial v
u
$f(u)$
v

RelaxNN: Macro-Micro Architecture & AP Loss

- **Chapman-Enskog Decoupling**: Micro-fluctuation $\mathbf{w}$ ($\mathbf{v} = \mathbf{f}(u) + \varepsilon \mathbf{w}$):

$$\begin{cases} \partial_t u + \nabla_x \cdot \mathbf{f}(u) = -\varepsilon \nabla_x \cdot \mathbf{w} \\ -\partial_t \mathbf{f}(u) - a \nabla_x u = \mathbf{w} + \varepsilon \partial_t \mathbf{w} \end{cases}$$

- **Asymptotic Preserving (AP) Loss ($\varepsilon \to 0$)**: Forces equilibrium manifold convergence:

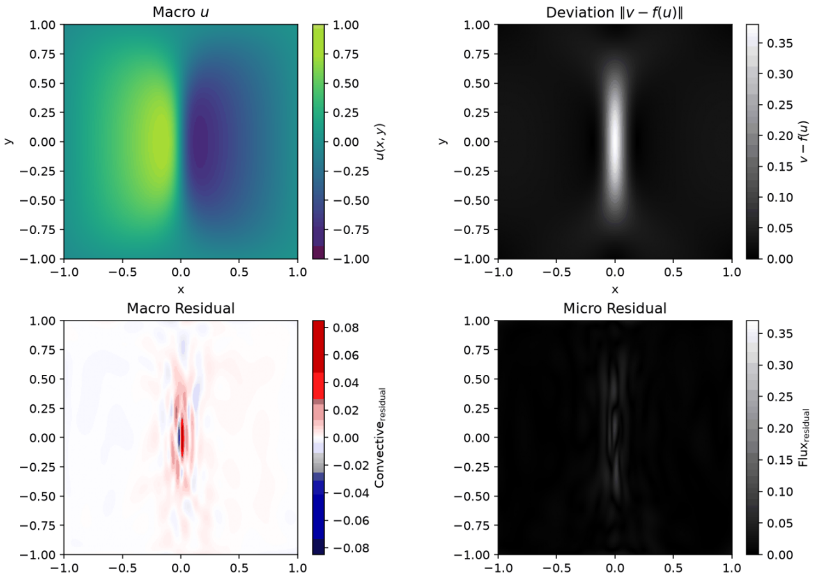
$$\mathcal{L}_0(\tilde{u}, \tilde{\mathbf{w}}) = \text{MSE}\left(\partial_t \tilde{u} + \nabla_x \cdot \mathbf{f}(\tilde{u}), 0\right) + \text{MSE}\left(-\partial_t \mathbf{f}(\tilde{u}) - a \nabla_x \tilde{u}, \tilde{\mathbf{w}}\right)$$

Remark: Decoupling acts as an ε-shield, avoiding high-frequency oscillations and reducing training time from ~ 300k to standard budgets.

- **Network Topology**: Shared deep **SiLU Encoder** branching into decoupled heads for macro ($\tilde{u}$) and micro ($\tilde{\mathbf{w}}$) fields.
- **Geometric Shock Shaping**: Final Tanh layer mimics analytical viscous shock:

$$u(\mathbf{x}, t) = \frac{u_L + u_R}{2} - \frac{u_L - u_R}{2} \tanh\left(\frac{(u_L - u_R)(\mathbf{x} - \mathbf{s}t)}{4\nu}\right)$$

- **Loss Formulation**: Combined MSE (boundaries/IC) and MAE $\mathcal{L}_1$ (residuals for sharp shock capturing) + L_1/L_2 regularizations.



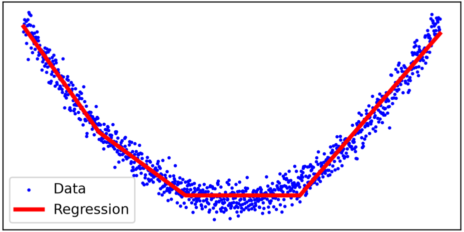
Macro u
Deviation $\|v - f(u)\|$
Macro Residual
Micro Residual
x
y
$u(x, y)$
$v - f(u)$
$\text{Convective}_{\text{residual}}$
$\text{Flux}_{\text{residual}}$

Implicit Explicit Neural Network (Proposal)

The Challenge: Sub-grid information loss on coarse grids. Classical stencils (Rusanov) trade off numerical diffusion vs unphysical oscillations.

Objective: Optimize *only* the discrete spatial operator via data, embedded in a conservative IMEX framework.

- Circumvents black-box temporal tracking.
- Operates as a physics-bounded spatial regression function.



Observed Pitfalls and Proposed Loss

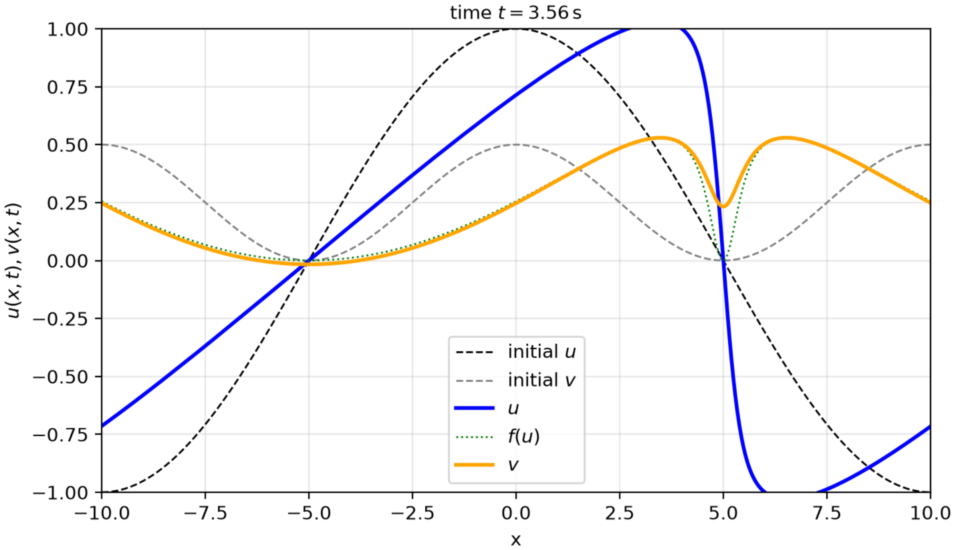
Stability Traps

- **Manifold Attraction**: Overfitting to physical diffusion instead of fluxes.
- **Central Bias**: FCNs default to non-dissipative symmetric stencils, triggering explosions.

Constrained Optimization Framework

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{interp}} + \lambda_1 \mathcal{L}_{\text{additivity}} + \lambda_2 \mathcal{L}_{\text{stability}}$$

- $\mathcal{L}_{\text{interp}}$: Local error vs reference gradients.
- $\mathcal{L}_{\text{additivity}}$: Semi-group coherence ($\Delta t/2 \times 2 = \Delta t$).
- $\mathcal{L}_{\text{stability}}$: Spectral radius control via Fourier Analysis.


time $t = 3.56\,\mathrm{s}$
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x
$u(x,t), v(x,t)$
initial u
initial v
u
$f(u)$
v

Expectations and Open Questions

Remark

The data-driven spatial operator is a structurally bounded optimization conditioned by training distribution.

Core Research Directions

- **Function Space Adaptation**: Family of profiles identified during training and ReLU activation mapping.
- **Shock Specialization**: Can non-linearities act as dynamic, data-driven flux limiters without manual tuning?

ADAMUS Project, funded by the Italian Ministry of University and Research (MUR) under Directorial Decree No. 1236 of Aug 1, 2023 - Italian Science Fund - Call FIS2, CUP F53C25000210001